

Digit Test Process

We chose Windows 98 SE as the default OS for the test. The latest drivers for the motherboard chipset and the ATA/100 controller on the motherboard were loaded, as were the drivers for the USB ports for testing the external USB drives. All the CD-Writers were tested with DMA enabled for all IDE channels. A ghost image of the current install was backed up and reinstated after testing each writer to ensure minimum drive fragmentation.

The CD-Writers under test were benchmarked for their performance as CD writers and for their CD reading capability. The latter consisted of a blend of synthetic tests that logged low-level parameters

such as data transfer rates and access times of the drive, while the former tested the time taken to write CDs at various speeds.

The media used was the HP CD-R 700 MB, specified for a maximum of 24x (Cyanine, Type 1). The CD-RW media used was the 650 MB HP CD-RW, specified for a maximum of 10x (Cyanine, Type 3).

Write Tests: Here we tested for ISO write speed, audio write speeds, RAW write speeds, RW-assorted speeds and other assorted write speeds wherein a combination of file sizes ranging from 1 KB to 100 MB were burnt. We checked whether the drive performed these operations reliably and if there were buffer underruns or other

errors during the CD creation process.

Read Tests: The read tests comprised the audio extraction test, Nero CD speed test and an assorted read test in which a combination of file sizes ranging from 1 KB to 100 MB were read from the writer and copied to the hard drive.

Features and Package Contents: We also checked for buffer size, burn formats supported, the ability to read/write RAW data, etc. Finally, we examined whether the writers came with comprehensive instruction manuals, quick reference cards, software CDs, free blank media and other accessories. We also tested the bundled free CD writing software for ease of use.

the really fast ones can even manage to touch the magical 4-minute barrier.

In this rather intensive test (primarily because the system reads and writes a relatively large file in one continuous sequence) the ASUS CRW-2410S turned out to be a scorcher, scoring a speedy 241 seconds flat! Yes, a 700 MB ISO image was burnt in 4.01 minutes. Tough competition was to follow soon. The Yamaha CRW 3200E-VK blazed in just 3 seconds later, followed by the Plextor PX-W2410TA which completed the writing in 247 seconds (4.07 minutes). A difference of less than 10 seconds separated the top three drives, showing all too clearly that all of them were top-notch performers worthy of being in the big league.

All the 24x drives showed some real performance here and the difference between them can only be attributed to better access and beam assembly movement times.

Assorted hurdle

A bunch of files with a bunch of folders with long file names—try as you might, organising your files with proper names and folder structures is simply asking for too much isn't it? But that should not give a CD-Writer enough excuse to write a coaster.

Assorted files have always been a big problem to deal with. You will on the odd occasion find a few corrupt files written or that odd zipped file that simply refuses to extract. So we made the weirdest combination of assorted files, bunched up in a maze of folders with long file names. An exact 650 MB was accumulated (a

Burn Proof Technology

Buffer underruns occur when the computer fails to provide the data to the CD-Writer in a continuous stream. As a result, the recording fails and the media becomes unusable. This can happen when attempting to burn a CD-R whilst

performing other tasks, or when recording from a relatively 'slow' source to a 'fast' target.

In the past, you needed to shut down all the other activities running on your PC and wait till the CD-Writer finished writing

the CD. With the advent of Burn Proof technology you no longer need to continuously monitor the buffer bar. Introduced first by Sanyo, CD-Writers featuring this technology have become intelligent enough to synchronise, wait and then burn the data.

Using this technology, recording is paused at a specific location when an impending buffer-underrun condition is detected. The writing process resumes only when the buffer has been sufficiently replenished and after repositioning the CD-Writer's optical pickup mechanism to the appropriate sector. (See diagram, 'Streaming Data'.)

Most major drive manufacturers either license similar technology or have developed their own variants. Mitsumi and LG Electronics use OAK Technology's ExacLink system, Yamaha does the same but under the brand name SafeBurn. Acer Communications refers to their technology as Seamless Link and Ricoh calls theirs JustLink.

